Research Methodology

BY
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Agenda

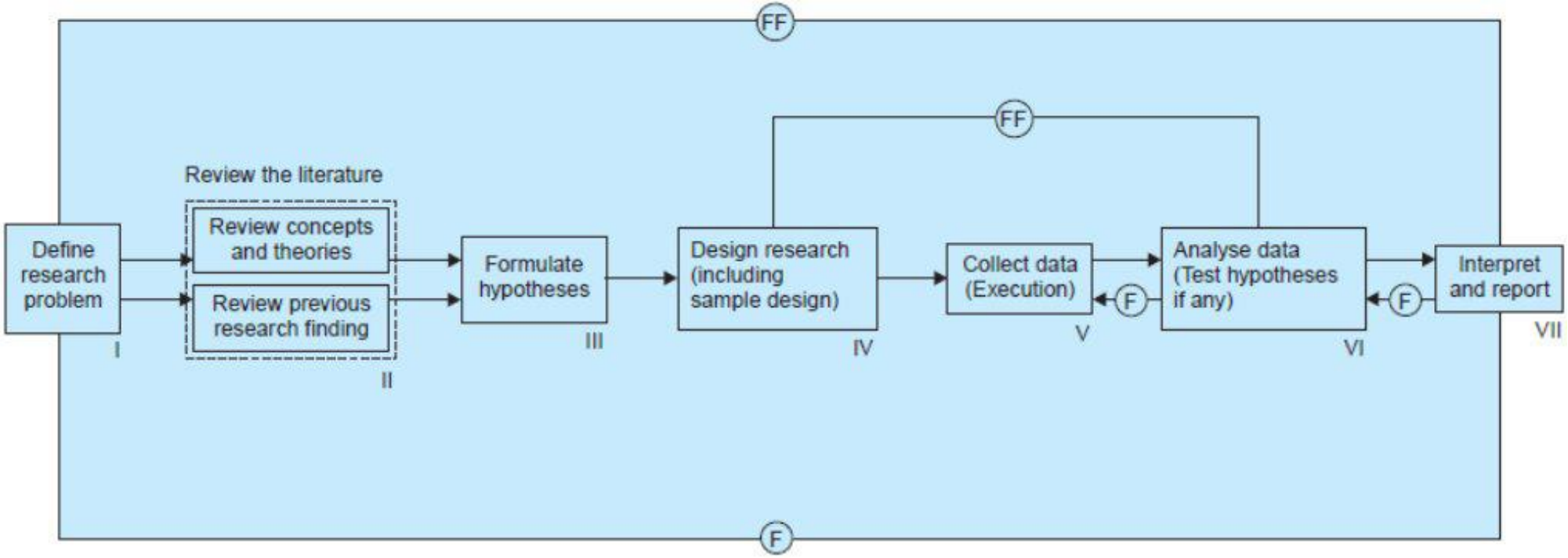
- Components of Research Methodology:
  - Research problem,
  - Objectives,
  - Research Design,
  - Data Collection,
  - Sampling Techniques,
  - Data Analysis,
  - Hypothesis Formulation,
  - Ethical Consideration and
  - Validation and Reliability
- Research Design: Experimental, Exploratory, Descriptive and Explanatory
- Research Strategies: Experimental Research, Case Study, Action Research, Theoretical Research, Simulation-based Research.
- Multidisciplinary Research Approaches

What is research?

- "Scientific research is systematic, controlled, empirical, and critical investigation of natural phenomena guided by theory and hypotheses about the presumed relations among such phenomena."
  - Kerlinger, 1986

- *Research is an organized and systematic way of finding answers to questions*

Research Process



Research Process

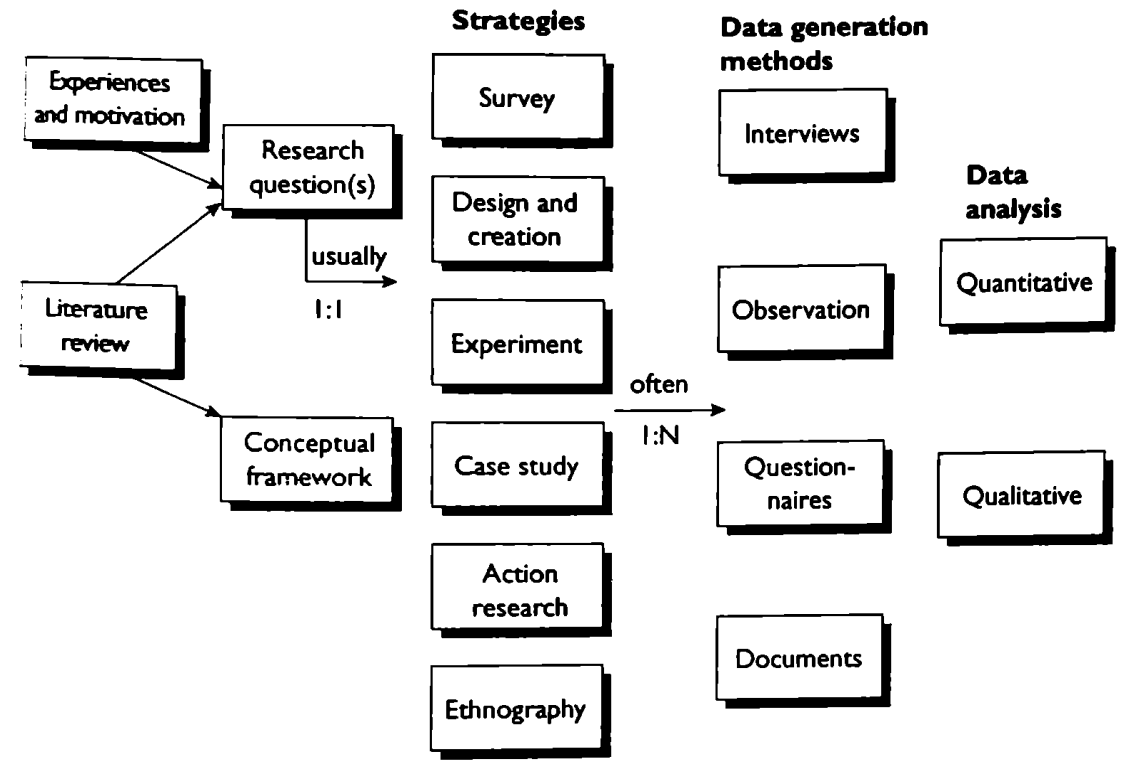


Figure 3.1 Model of the research process

Research problem

The key effort is to make the reviewers/readers understand

- What is the problem
- Why is the problem important

Problem statement answers one primary question: *Why* do this project?

- The emphasis is on the "why"

To answer this question, researcher must

- (1) define the problem and
- (2) document its existence

Research problem

The problem statement of the problem should quickly:

- summarize the problem from the vantage point of the readers,
- document its frequency and severity,
- show your familiarity with prior research or work on the problem, and
- justify why this problem should be investigated

Do not assume that everyone sees the problem as clearly as you do

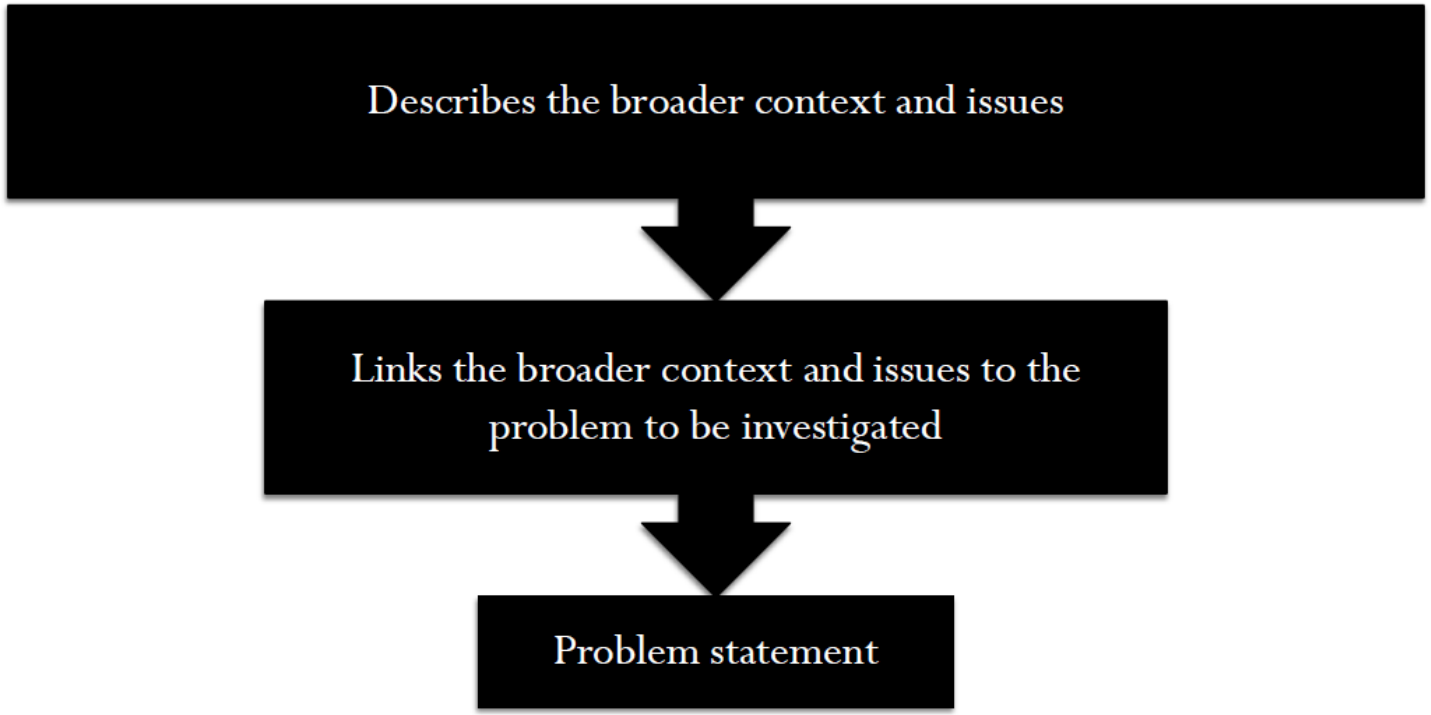
- even if the problem is obvious, your readers want to know how clearly you can state it

1.2 Problem Statement

The focus of this thesis is on introducing network services rather than using or adapting existing network services. Introducing network services is no new activity, this can already be done with a GPL. But since every technically possible problem can be solved with a GPL, there is no clear notion of the mechanisms that are required to introduce network services. It is expected that by developing a DSL, it will become clear which mechanisms are essential or common to introduce network services.

Let's Evaluate

Research problem



Research problem

Research is extending the knowledge frontier

- That is, building upon other researchers‟ work

Cite other people's work (not only yours)

Explain

- The current limits and gaps in knowledge
- Your study's contribution to fill in current gaps in knowledge or to extend the knowledge frontier
- The broader impact or benefit to the society, country, or the world
- A good statement to have in your research: *"If successful, the benefits of this research would be …"*

Objectives

Objectives describe the purpose of the study

Must be
- specific
- clear
- logical
- immediate (have a time frame)
- concise, and
- achievable (measurable)

Anticipate	Construct	Discriminate	Measure
Arrange	Contrast	Display	Motivate
Assemble	Coordinate	Distinguish	Organize
Assess	Decrease	Establish	Quantify
Build	Demonstrate	Estimate	Solve
Categorize	Describe	Evaluate	Stimulate
Classify	Design	Explain	Summarize
Compare	Detect	Illustrate	Translate
Conduct	Discover	Increase	

May have two sections: general and specific objectives

Or split study into parts/sections, and for each section a statement of its study objectives
- But ensure the sections are related to one another to solve or investigate the overall problem

Hypothesis to be tested
- Expected results
- What you expect/believe the results would be

Use action verbs in objectives such as:

Research Design

Definitions

- 'as the arrangement of conditions for collection and analysis of data in a manner that aims to combine the relevance to the **research purpose with economy in procedure**'-*by Jahoda*
- 'a master plan, specifying the methods and procedures for collecting and analyzing the needed information'-*by Zikmund*
- A research design is a plan, structure and strategy of investigation so conceived as to obtain answers to research questions or problems. It includes an outline of what the investigator will do from writing the hypotheses and their operational implications to the final analysis of data. (Kerlinger, 1986)
- 'A research design is the arrangement of conditions for collection and analysis of data in a manner that aims to combine relevance to the research purpose with economy in procedure'. According to Selltiz, Deutsch and Cook(1962),

Research designs are the specific procedures involved in the research process: data collection, data analysis, and report writing.

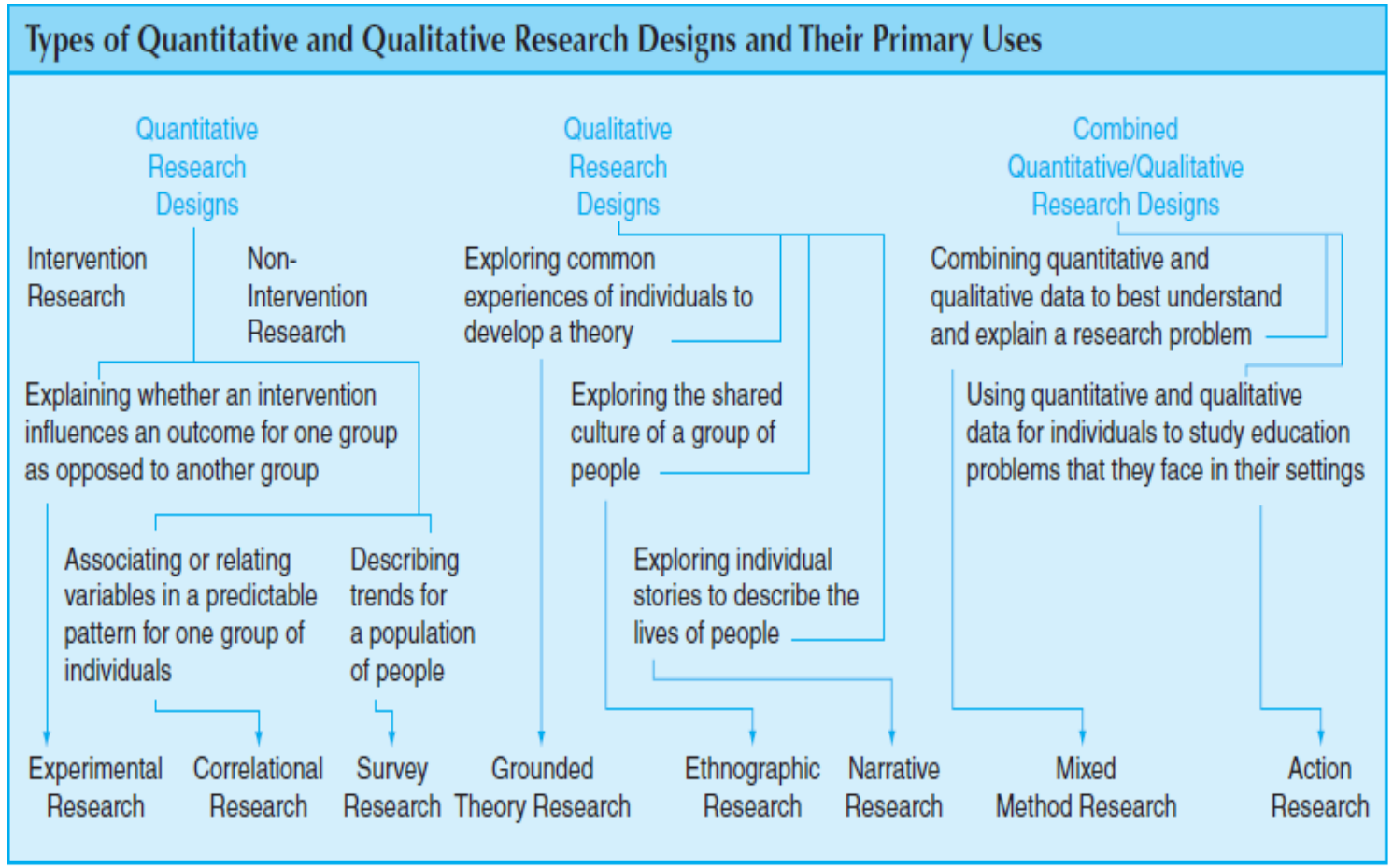
The term "research design" means 'drawing for research'.

It is a systematic planning of conducting research.

It aims to achieve goals of the research.

Research Designs

Associated with Quantitative and Qualitative Research



Data Collection

Need for Data Collection

- Foundation of research – raw material for analysis.
- Provides direction and clarity to research inquiries.
- Verifies and tests hypotheses for scientific rigor.
- Substantiates findings with empirical evidence.
- Enables estimation and generalization of results.
- Combines qualitative (facts) and quantitative (theories) insights.
- Ensures practical utility and problem-solving in research.

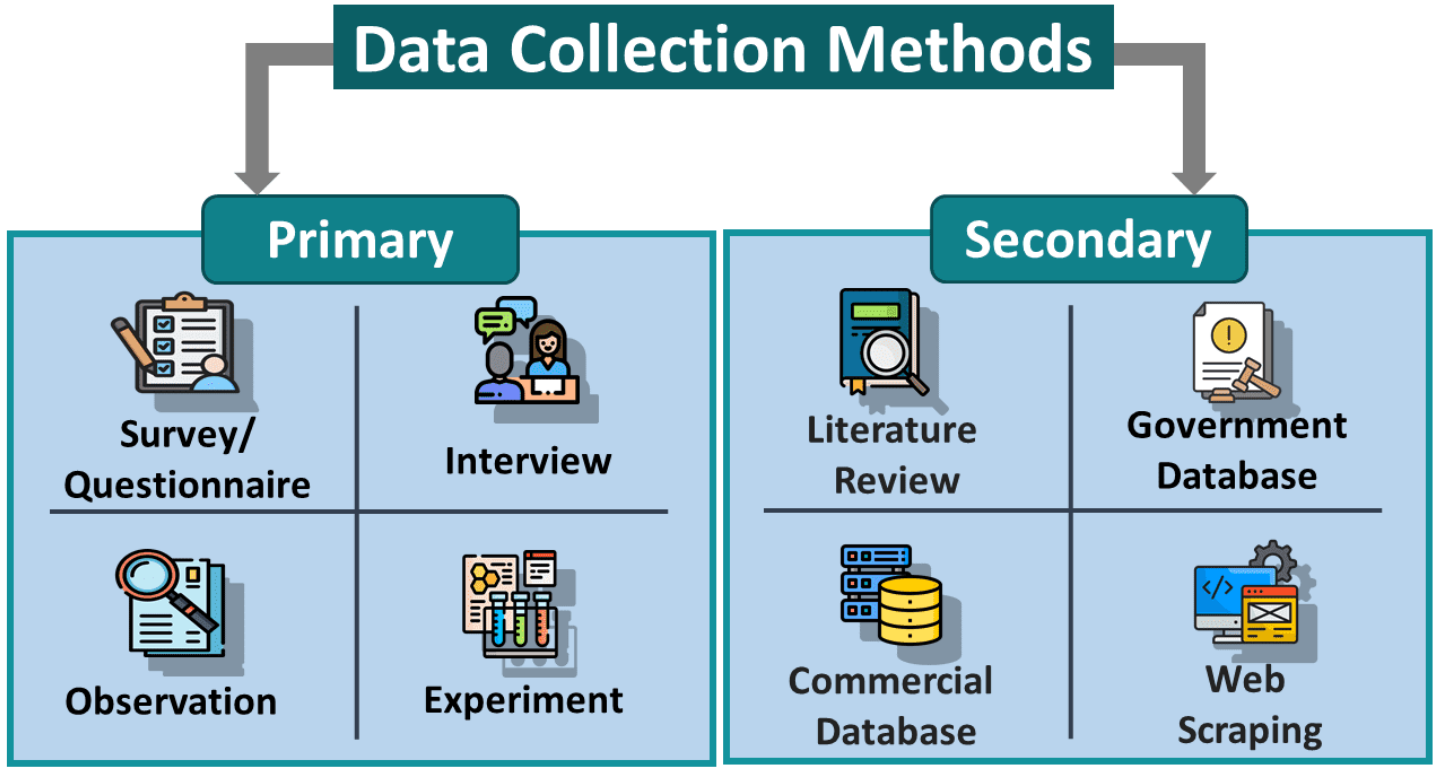
Data Collection Methods

Interviews

Observations

Questionnaires

Focus groups



Sampling Techniques

A sample is “a smaller (but hopefully representative) collection of units from a population used to determine truths about that population” (Field, 2005)

Why sample?

- Resources (time, money) and workload
- Gives results with known accuracy that can be calculated mathematically

The sampling frame is the list from which the potential respondents are drawn

- Registrar’s office
- Class rosters
- Must assess sampling frame errors

SAMPLING……

What is your population of interest?

- To whom do you want to generalize your results?
  - All IT Graduates
  - School children
  - Indians
  - Women aged 15-45 years
  - Other

Can you sample the entire population?

3 factors that influence sample representative-ness

- Sampling procedure
- Sample size
- Participation (response)

When might you sample the entire population?

- When your population is very small
- When you have extensive resources
- When you don’t expect a very high response

Errors of nonobservation

The deviation between an estimate from an ideal sample and the true population value is the sampling error.

Almost always, the sampling frame does not match up perfectly with the target population, leading to errors of coverage.

Errors of nonobservation

Nonresponse is probably the most serious of these errors.

- Arises in three ways:
  - Inability of the person responding to come up with the answer
  - Refusal to answer
  - Inability to contact the sampled elements

Errors of observation

These errors can be classified as due to the interviewer, respondent, instrument, or method of data collection.

Interviewers

Interviewers have a direct and dramatic effect on the way a person responds to a question.

- Most people tend to side with the view apparently favored by the interviewer, especially if they are neutral.
- Friendly interviewers are more successful.
- In general, interviewers of the same gender, racial, and ethnic groups as those being interviewed are slightly more successful.

Respondents

Respondents differ greatly in motivation to answer correctly and in ability to do so.

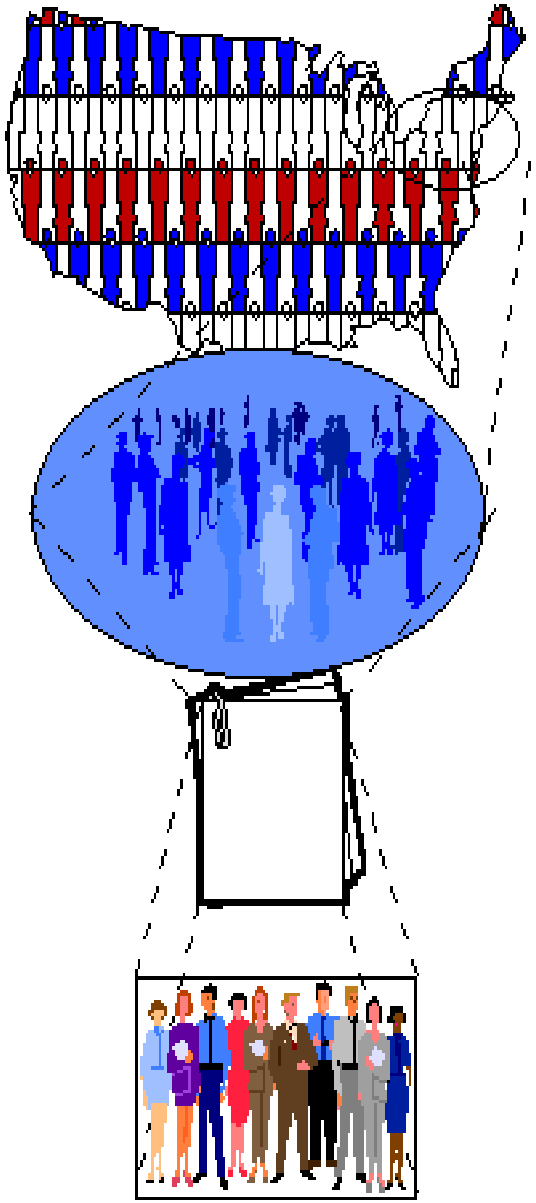
Obtaining an honest response to sensitive questions is difficult.

Basic errors

- Recall bias: simply does not remember
- Prestige bias: exaggerates to ‘look’ better
- Intentional deception: lying
- Incorrect measurement: does not understand the units or definition

Who do you want to generalize to?

The Theoretical Population



What population can you get access to?

The Study Population

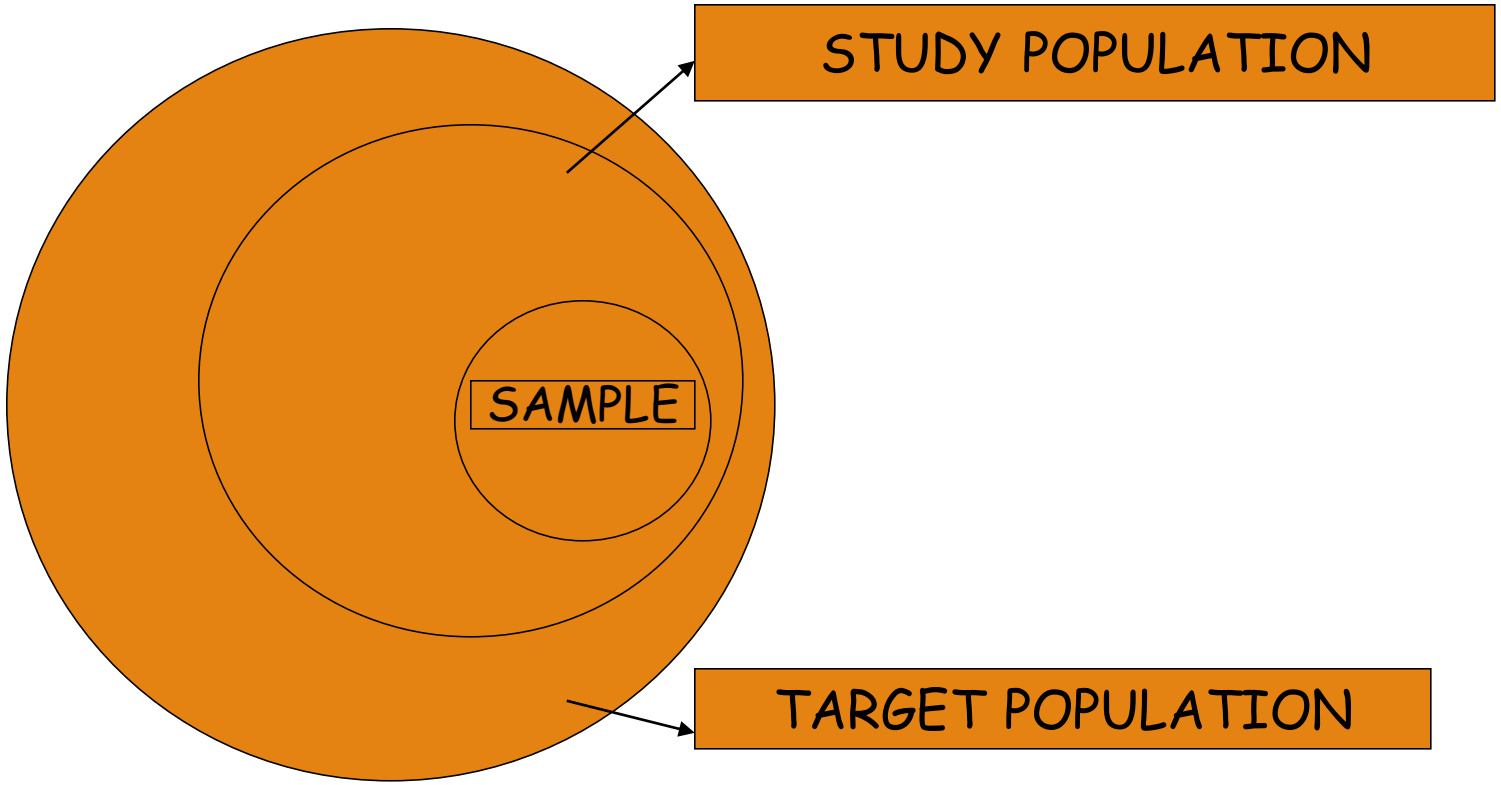
How can you get access to them?

The Sampling Frame

Who is in your study?

The Sample

SAMPLING.....



Types of Samples

Probability (Random) Samples

Simple random sample

- Systematic random sample
- Stratified random sample
- Multistage sample
- Multiphase sample
- Cluster sample

Non-Probability Samples

- Convenience sample
- Purposive sample
- Quota

Process

The sampling process comprises several stages:

- Defining the population of concern
- Specifying a sampling frame, a set of items or events possible to measure
- Specifying a sampling method for selecting items or events from the frame
- Determining the sample size
- Implementing the sampling plan
- Sampling and data collecting
- Reviewing the sampling process

PROBABILITY SAMPLING

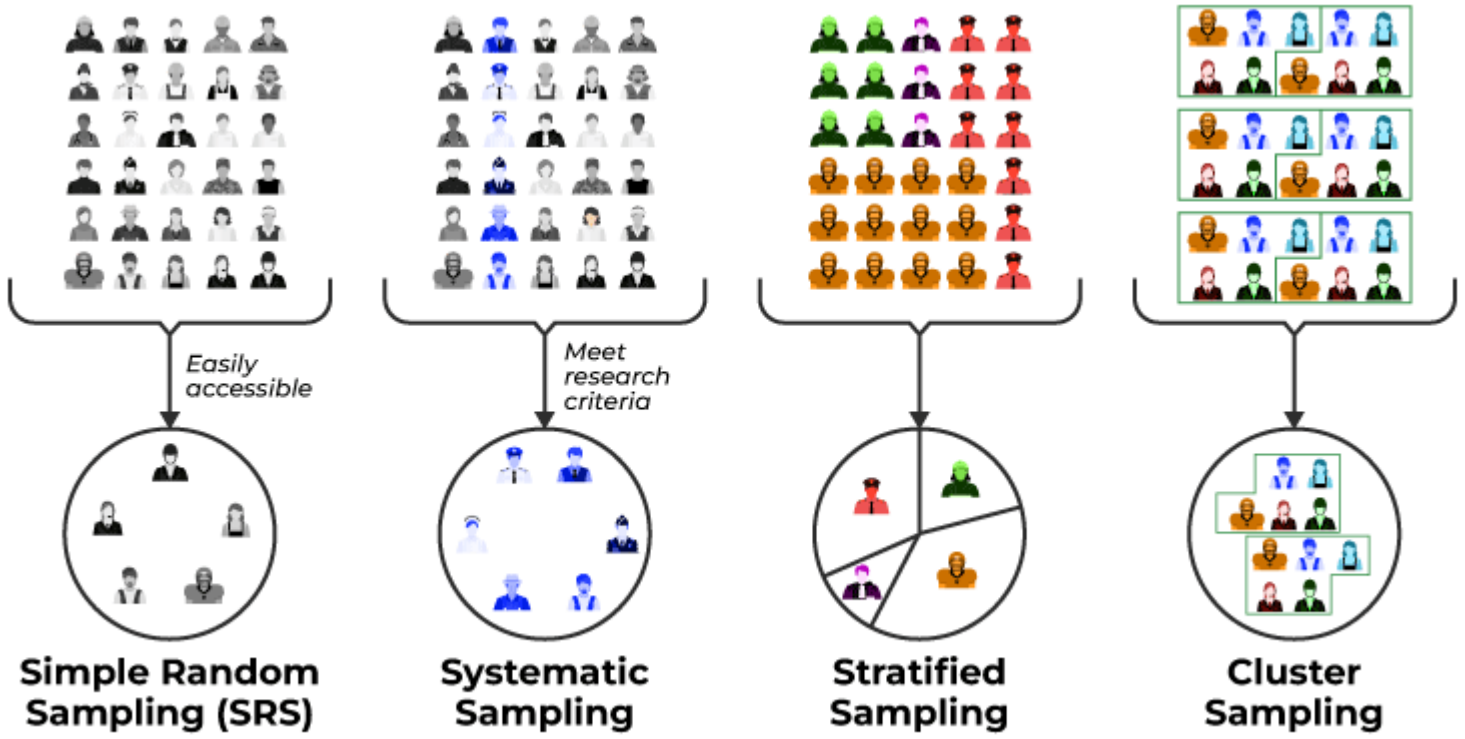
A **probability sampling** scheme is one in which every unit in the population has a chance (greater than zero) of being selected in the sample, and this probability can be accurately determined.

. When every element in the population *does* have the same probability of selection, this is known as an 'equal probability of selection' (EPS) design. Such designs are also referred to as 'self-weighting' because all sampled units are given the same weight.

PROBABILITY SAMPLING…….

Probability sampling includes:

- Simple Random Sampling,
- Systematic Sampling,
- Stratified Random Sampling,
- Cluster Sampling



Random Sampling

Random sampling is the purest form of probability sampling.

Each member of the population has an equal and known chance of being selected.

When there are very large populations, it is often ‘difficult’ to identify every member of the population, so the pool of available subjects becomes biased.

- You can use software, such as minitab to generate random numbers or to draw directly from the columns

Systematic Sampling

Systematic sampling is often used instead of random sampling. It is also called an Nth name selection technique.

After the required sample size has been calculated, every Nth record is selected from a list of population members.

As long as the list does not contain any hidden order, this sampling method is as good as the random sampling method.

Its only advantage over the random sampling technique is simplicity (and possibly cost effectiveness).

Stratified Sampling

Stratified sampling is commonly used probability method that is superior to random sampling because it reduces sampling error.

A stratum is a subset of the population that share at least one common characteristic; such as males and females.

- Identify relevant stratums and their actual representation in the population.
- Random sampling is then used to select a *sufficient* number of subjects from each stratum.
- Stratified sampling is often used when one or more of the stratums in the population have a low incidence relative to the other stratums.

Cluster Sampling

Cluster Sample: a probability sample in which each sampling unit is a collection of elements.

Effective under the following conditions:

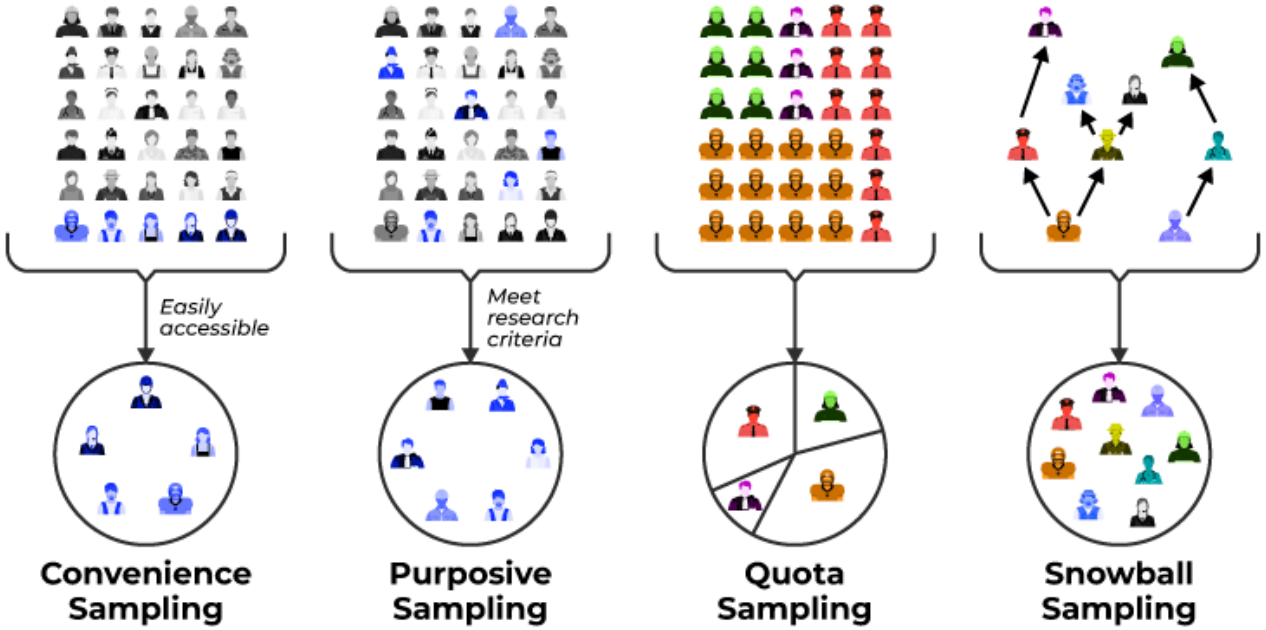
- A good sampling frame is not available or costly, while a frame listing clusters is easily obtained
- The cost of obtaining observations increases as the distance separating the elements increases

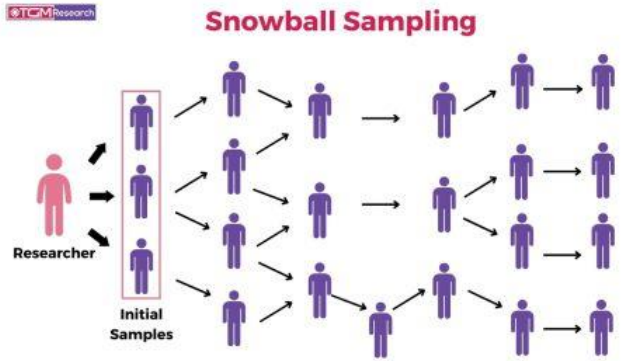
Examples of clusters:

- City blocks – political or geographical
- Housing units – college students
- Hospitals – illnesses
- Automobile – set of four tires

NONPROBABILITY SAMPLING……

- Nonprobability Sampling includes: Accidental Sampling, Quota Sampling and Purposive Sampling. In addition, nonresponse effects may turn *any* probability design into a nonprobability design if the characteristics of nonresponse are not well understood, since nonresponse effectively modifies each element's probability of being sampled.





Convenience Sampling

Convenience sampling is used in exploratory research where the researcher is interested in getting an inexpensive approximation.

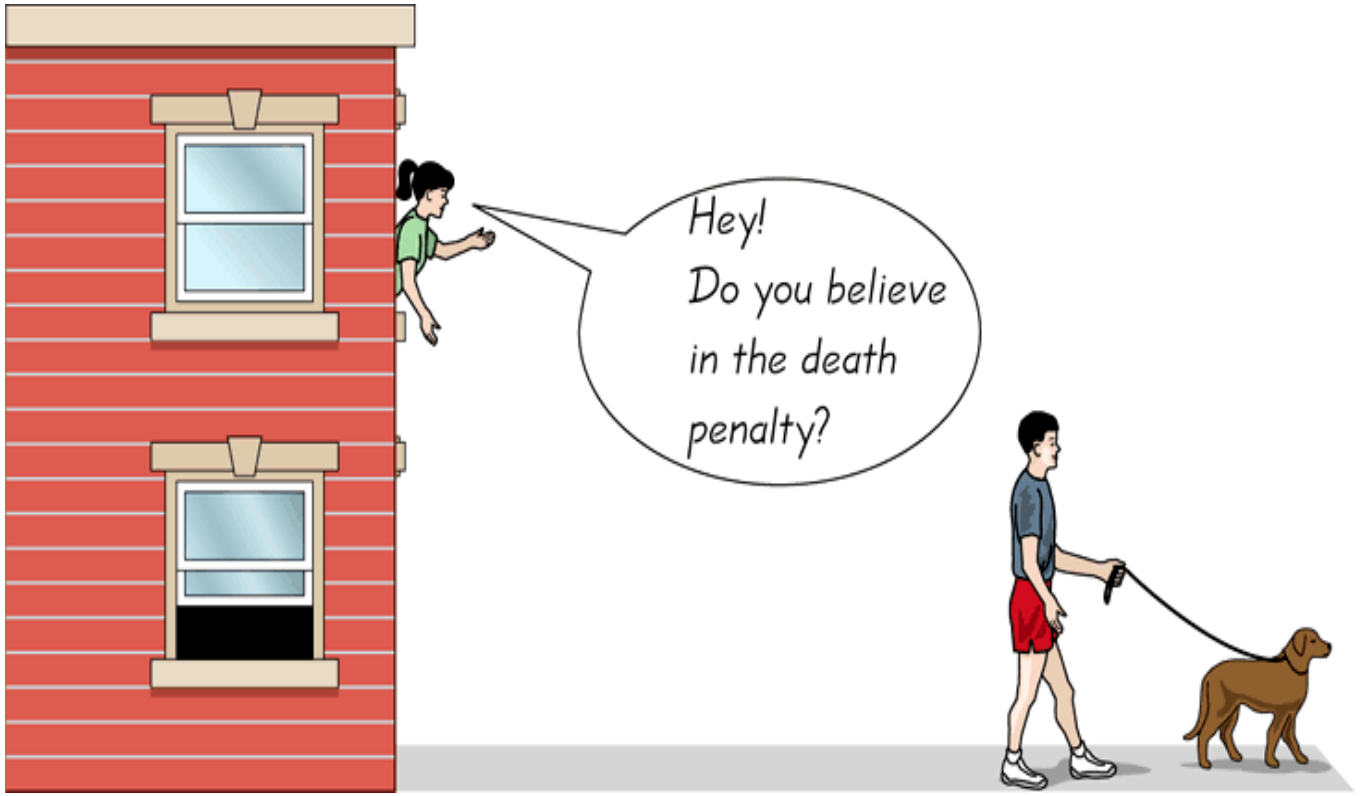
The sample is selected because they are convenient.

It is a nonprobability method.

- Often used during preliminary research efforts to get an estimate without incurring the cost or time required to select a random sample

CONVENIENCE SAMPLING…….

- Use results that are easy to get



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Judgment Sampling

Judgment sampling is a common nonprobability method.

The sample is selected based upon judgment.

- an extension of convenience sampling

When using this method, the researcher must be confident that the chosen sample is truly representative of the entire population.

Quota Sampling

Quota sampling is the nonprobability equivalent of stratified sampling.

- First identify the stratums and their proportions as they are represented in the population
- Then convenience or judgment sampling is used to select the required number of subjects from each stratum.

Snowball Sampling

Snowball sampling is a special nonprobability method used when the desired sample characteristic is rare.

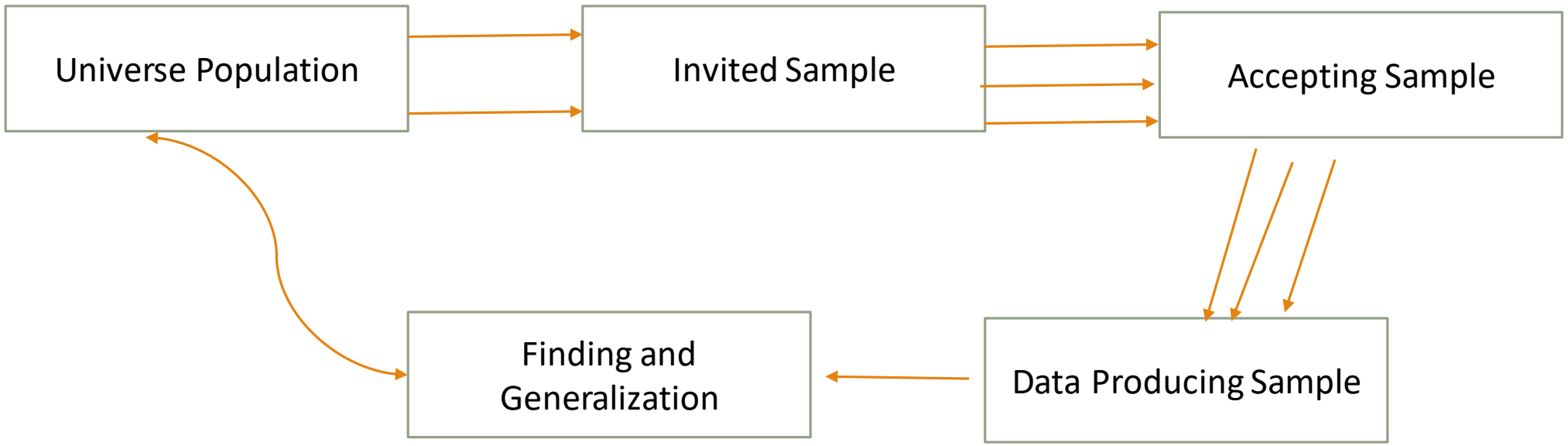
It may be extremely difficult or cost prohibitive to locate respondents in these situations.

This technique relies on referrals from initial subjects to generate additional subjects.

It lowers search costs; however, it introduces bias because the technique itself reduces the likelihood that the sample will represent a good cross section from the population.

THE SAMPLING CYCLE

Five stages of sampling cycle are proposed here, as sketched in figure.



The Sample Cycle

Data Analysis

- Quantitative Data Analysis
  - Data Presentation: Tables, Bar charts, Scatter Graph, Line graph
  - Analysis Techniques: Descriptive Statistics, Inferential Statistics (Z-test, t-test, Chi-square test, ANOVA), Regression Analysis, Time Series Analysis and Multivariate Analysis
- Qualitative Data Analysis

Hypothesis Formulation

Experiment

- Investigates cause-and-effect relationships.
- Aims to prove or disprove causal links between factors and outcomes.

Process of Developing a Hypothesis:

- Researchers begin with a theory on a topic of interest.
- Formulate a testable statement based on the theory.
- Hypothesis format: "Factor A causes B."

Examples of Hypotheses:

- The Watson-Klein-Bloggs algorithm enables faster data processing than the current algorithm.
- Smoking cigarettes causes lung cancer.
- Students pay more attention in lectures without handouts.

Hypotheses Formulation

An experiment is based on a hypothesis to be tested.

A hypothesis is a statement that has not yet been tested empirically (that is, by gathering field data) but for which it is possible to devise empirical tests that will provide clear evidence to support it or reject it.

It is written as a kind of prediction, for example:

- When factor A occurs B will happen.
- An increase in D causes a decrease in C.
- Water will freeze when the temperature drops to 32 degrees Fahrenheit (0 degrees centigrade).

Ethical Consideration

Ethics may be defined as:

- Sets of moral principles or norms that are used to guide moral choices of behavior and relationships with others. (Blumberg, et al, 2005)
- The moral principles and actions guiding and shaping research. (ESRC, 2004)

Ethics in research - - The set of values, standards, and principles used to distinguish between appropriate and acceptable conduct and unacceptable conduct at all stages of the research process.

Introduction to Research Ethics

- Research ethics = norms for acceptable/unacceptable behavior in research.
- Ensures integrity, quality, and accountability in research.
- Ethical principles guide research design, conduct, and publication.

"Ethics is knowing the difference between what you have a right to do and what is right to do." – Justice Potter Stewart



Foundation of Shared Values

TRUST— Core ethical value in the scientific pursuit of the truth

HONESTY — Conveying information truthfully and honoring commitments

ACCURACY— Reporting findings precisely and taking care to avoid errors

OBJECTIVITY— Letting the facts speak for themselves and avoiding bias

Requisite for Ethics in Research

Nuremberg Code (1947) and Declaration of Helsinki (1994) set ethical standards.

- Principles: Honesty, integrity, protection of human subjects, respect for IPR, objectivity, and data management.
- Avoid plagiarism, ensure confidentiality, and promote openness.

Believing what you read?

". . . there is a need for reliable and valid data on student learning outcomes."

"Validity concerns the degree to which inferences about students based on their test scores are warranted."

-Cameron, L, SL Wise, and SM Lottridge. 2007. The Development and Validation of the Information Literacy Test. *College and research libraries* 68 (3):229.

Reliable and Valid Data

"The team's combined goal was to produce a valid, reliable, authentic assessment of ICT literacy skills."

"The goal of the iSkills assessment is to measure the ICT literacy skills of students—higher scores on the assessment should reflect stronger skills."

-Katz, IR. 2007. Testing Information Literacy in Digital Environments: ETS's iSkills Assessment. *Information technology and libraries* 26 (3):3.

Reliability

In statistics or measurement theory, a measurement or test is considered reliable if it produces consistent results over repeated testings.

Refers to “how well we are measuring whatever it is that is being measured (regardless of whether or not it is the right quantity to measure).”

-D. Rindskopf, Reliability: Measurement. In: Neil J. Smelser and Paul B. Baltes, Editor(s)-in-Chief, International Encyclopedia of the Social & Behavioral Sciences, Pergamon, Oxford, 2001, Pages 13023-13028.
(http://www.sciencedirect.com/science/article/B7MRM-4MT09VJ2XN/1/083e3cc0b8b9d4e027b0ba214dcd9fa3)

Reliability

Unlike the common understanding, in these contexts “reliability” does not imply a value judgment

- Your car always starts/doesn’t start
- Your friend is always/ never late

Classical Test Theory (CTT)

A single trait or skill is being measured

The trait or skill can be defined

All items on the test measure the same trait or skill

Formula for determining reliability

Test is made more reliable by making it longer

Limitation: reliability depends upon the sample group and is “not a characteristic of the test itself.”

Generalizability Theory (GT)

Based on analysis of variance

Unlike CTT, GT allows for multiple sources of error

The test is designed to account for factors that researchers predict will influence scores

Can compute multiple estimates of reliability

Item Response Theory (IRT)

Like CCT, IRT measures a single trait or skill

Relationship between the score on an individual test item and the skill/trait can be measured

“Adaptive tests” – tests can be customized to the individual test-taker, e.g., the GRE

Does not use the traditional concept of reliability

Observational Studies

Some characteristics cannot be measured through a test

Unobtrusiveness

Multiple sources of error

Reliability depends on the extent to which observers agree

Validity Evidence

Content Validity: “that based on expert ratings of the items” in the test

Construct Validity: “that based on the degree to which ILT scores statistically behave as we would expect a measure of information literacy to behave.”

- Cameron, L, SL Wise, and SM Lottridge. 2007. The Development and Validation of the Information Literacy Test. *College and research libraries* 68 (3):229.

How can validity be established?

Quantitative studies:

- measurements, scores, instruments used, research design

Qualitative studies:

- ways that researchers have devised to establish credibility: member checking, triangulation, thick description, peer reviews, external audits

How can reliability be established?

Quantitative studies?

- Assumption of repeatability

Qualitative studies?

- Reframe as dependability and confirmability

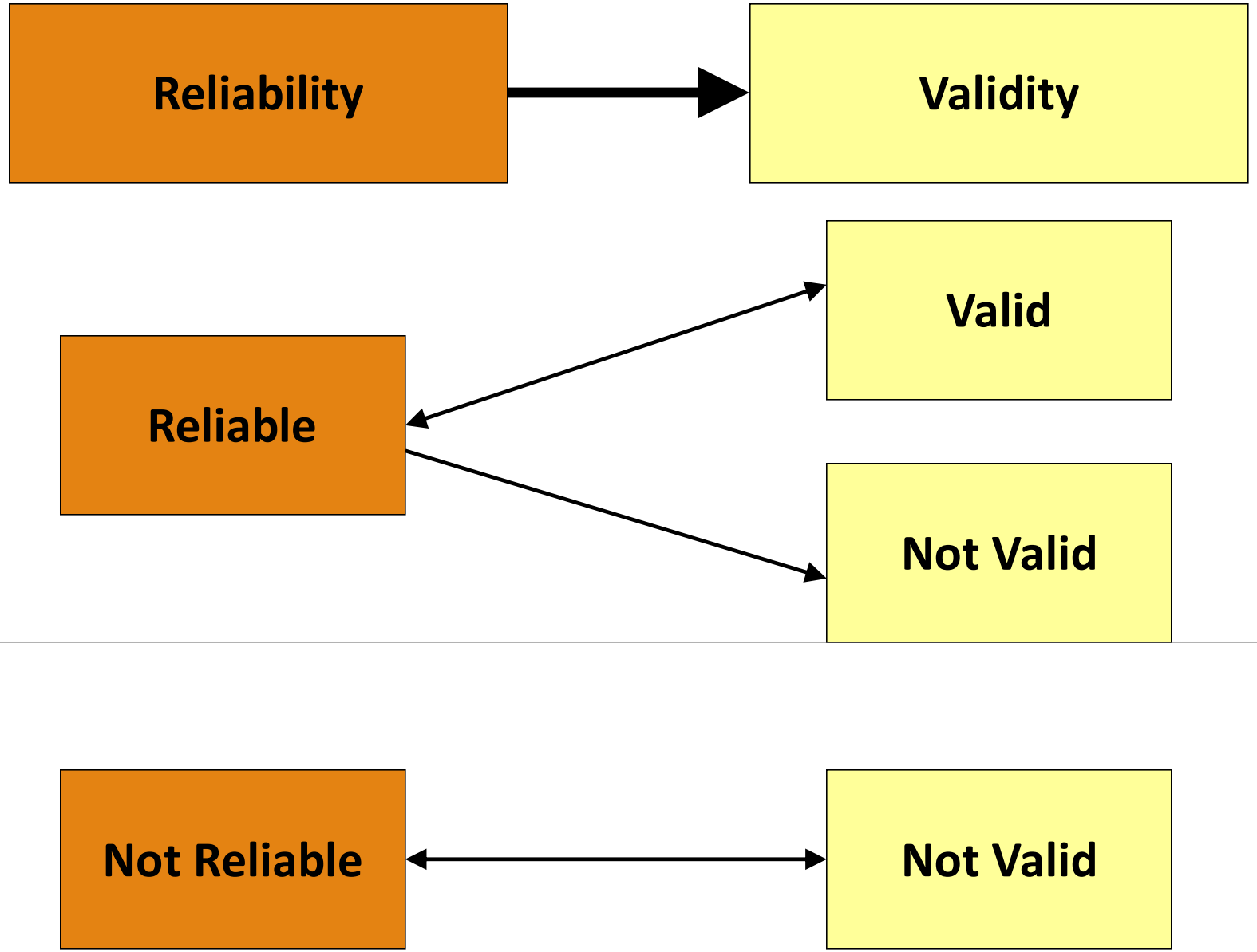
Reliability and Validity

Why do we bother?

Terms used in conjunction with one another

- Quantitative Research: R & V are treated as separate terms
- Qualitative Research: R & V are often all under another, all encompassing term

Semi-reciprocal relationship


Reliability
Validity
Reliable
Valid
Not Valid
Not Reliable
Not Valid

Research Design

A research design is a **plan** according to which observations are made and data assembled. It provides the empirical and logical basis for drawing conclusions and gaining knowledge.

Why?

- Provides the blueprint for research
- limits the boundaries f research activity
- enables investigation to anticipate potential problems

  (Balck and Champion, 1976)

Types

- Experimental
- Exploratory
- Descriptive and
- Explanatory

Ahuja (2007) explains that the research designing proceeds in eight phases as under:

1) **Stating the problem**, i.e., to show the type of research it will be, and to what extent will it contribute to the academics and understanding of society.
2) **Review of other studies**, i.e., going through the findings, hypotheses or theories developed by other scholars in one's own discipline or other disciplines.
3) **Operationaliing concepts**, i.e., giving specific meanings to the terms used.
4) **Identifying variables of the study**, i.e., pointing out key variables (see BOX) in the study and methods of measurement.
5) **Fixing sample**, i.e., deciding the numbers of subjects from whom the data is to be collected and how these subjects are to be chosen.
6) **Specifying tools of study**, i.e., whether data are to be obtained through questionnaires, schedule, interview or observation. Whether it will be case study, survey study, field study or experimental study.
7) **Designing the kind of analysis**, i.e., whether quantitative or qualitative analysis for the study.
8) **Fixing time-schedule and budget**, i.e., how much time and budget is to be devoted to the study.

Descriptive Design

Descriptive research studies aim to explain the features of a particular individual, group, or situation.

Focuses on describing **what is** rather than **why it is**.

Key Purpose:

- To provide a detailed and accurate interpretation of a phenomenon.
- Often used in social research to describe characteristics or behaviors

The descriptive designs enable researchers to describe or presents picture of the phenomenon or phenomena under investigation.

The methodology involved in such designs is mostly qualitative in nature producing descriptive data i.e. people's own written or spoken words and observable behavior.

There are at least three such approaches, which enable us to record or analyze the behavioral patterns.

- Participant Observation
- Personal Documents, and
- Unstructured Interviewing

Characteristics of Descriptive Research Design

Objective:
- To describe the current state of a phenomenon.

Focus:
- Narration of facts, characteristics, and specific predictions.

Design Requirements:
- Exact and thorough: The design must be precise to avoid bias.
- Shield against bias: Ensures reliability and validity of data.
- Efficient conclusion: Focuses on achieving clear and actionable results.

Types of Descriptive Research
- Descriptive Studies:
  - Focus on describing the characteristics of a group or phenomenon.
  - Example: A study on the demographic profile of a city.
- Diagnostic Studies:
  - Aim to determine the frequency of an occurrence or its relationship with other variables.
  - Example: A study on the relationship between smoking and lung cancer.

Steps in Descriptive Research Design

Identify Objectives:
- Clearly define what the study aims to achieve.

Design Data Collection Methods:
- Choose appropriate techniques (e.g., surveys, interviews, observations).

Select Sample:
- Determine the sample size and selection criteria.

Collect Data:
- Gather data from reliable sources within a specific time frame.

Manage and Evaluate Data:
- Organize and analyze the collected data.

Report Outcomes:
- Present findings in a structured and comprehensible manner.

Exploratory Method

The exploratory method represents the earlier stage of science (According to Katz).

It implies that all sciences must have at the beginning had an approach, which was purely exploratory.

Even in practical life, if we go to a doctor for some treatment, he systematically explores the complaints and tries to categories our symptoms.

On the basis of this exploration, he arrives at a conclusion, at least, tentatively, about the disease.

All these steps, starting with exploratory questions helps him or her to arrive at correct diagnosis.

Exploratory Research Method

Exploratory research is used to investigate a problem that is not clearly defined.

It helps in **formulating problems** and developing **hypotheses** for further research.

Purpose:

- To gain insights and understanding of a phenomenon.
- To explore new ideas, concepts, or trends.

Common Use Cases:

- When the research problem is vague or poorly understood.
- In the early stages of research to identify key variables and relationships.

Methods Used in Exploratory Research

Survey of Relevant Literature:

- Reviewing existing studies to identify gaps and formulate hypotheses.
- Helps in building a theoretical foundation for the research.

Experience Survey:

- Consulting experts, professionals, or individuals with firsthand knowledge of the problem.
- Provides practical insights and guidance for the study.

Analysis of Insight-Stimulating Examples:

- Studying specific cases or examples to generate new ideas or hypotheses.
- Useful when there is little prior research or experience to guide the study.

Experimental Research Design

Experimental research design investigates **cause-and-effect relationships** by manipulating one variable (independent variable) to observe its effect on another (dependent variable).

Key Features:

- Involves **controlled experiments** with an experimental group and a control group.
- Aims to establish **causal relationships** between variables.

Common Use Cases:

- Used in fields like **social psychology**, **organizational studies**, and **policy evaluation**.
- Example: Testing the impact of a new teaching method on student performance.

Types of Experimental Research Designs

After-Only Experiment:

- Two groups (experimental and control) are compared after the experimental group receives treatment.
- Limitation: Difficult to determine if differences are due to the treatment or external factors.

Before-After Experiment:

- Observations are made before and after the experiment in both groups.
- Advantage: Helps measure changes caused by the treatment.
- Limitation: External factors during the experiment may influence results.

Ex-Post Facto Experiment:

- Studies events that have already occurred (e.g., historical data).
- Example: Analyzing why a revolution occurred in one country but not in another.

Explanatory Research

Explanatory research design focuses on explaining why and how certain phenomena occur.

It aims to identify causal relationships between variables.

Purpose:

- To provide a deeper understanding of the underlying causes of a problem or phenomenon.
- Often used to test hypotheses and theories.

Common Use Cases:

- Used in social sciences, policy research, and market studies.
- Example: Investigating the factors that influence employee productivity.

Key Features of Explanatory Research Design

Focus on Causality:

- Seeks to establish cause-and-effect relationships between variables.
- Example: Does increased training (cause) lead to higher productivity (effect)?

Structured Approach:

- Uses rigorous methods such as experiments, surveys, or statistical analysis.
- Ensures reliability and validity of findings.

Hypothesis Testing:

- Involves formulating and testing hypotheses to explain observed phenomena.
- Example: Testing whether a new policy reduces unemployment rates.

Use of Control Groups:

- Often includes control groups to compare outcomes and isolate the effect of the independent variable.

	Exploratory Research	Descriptive Research	Explanatory Research
Degree of Problem Definition	Key variables not define	Key variables not define	Key variables not define
Researchable issue example	"The quality of service is declining and we don't know why."	"What have been the trends in organizational downsizing over the past ten years?"	"Which of two training programs is more effective for reducing labour turnover?
Researchable issue example	"Would people be interested in our new product idea?	"Did last year's product recall have an impact on our company's share price?"	"Can I predict the value of energy stocks if I know the current dividends and growth rates of dividends?"
Researchable issue example	"How important is business process reengineering as a strategy?	"Has the average merger rate for financial institutions increased in the past decade?"	"Do buyers prefer our product in a new package?"

Table 3.1. Exploratory, descriptive and explanatory research differences (Adapted from Adjei, n.d.)

Research Strategie

Experimental Research

Case Study

Action Research

Theoretical Research

Simulation-based Research

TABLE 6.3

Five Research Strategies Organized by the Data Structures They Use

Category 1: Strategies that examine individual variables.

Descriptive

Purpose: Produce a description of individual variables as they exist within a specific group.

Data: A list of scores obtained by measuring each individual in the group being studied.

Example: On average, students at the local college spend 12.5 hours studying outside of class each week and get 7.2 hours of sleep each night.

Category 2: Strategies that examine relationships between variables by measuring two (or more) variables for each participant.

Correlational

Purpose: Produce a description of the relationship between two variables but do not attempt to explain the relationship.

Data: Measure two variables (two scores) for each individual in the group being studied (see Figure 6.2).

Example: There is a relationship between wake-up times and grade point averages for college students, but we don't know why.

Category 3: Strategies that examine relationships between variables by comparing two (or more) groups of scores.

Experimental

Purpose: Produce a cause-and-effect explanation for the relationship between two variables.

Data: Create two treatment conditions by changing the level of one variable. Then measure a second variable for the participants in each condition (see Table 6.2a).

Example: Increasing the amount of exercise causes a decrease in cholesterol levels.

Quasi-Experimental

Purpose: Attempt to produce a cause-and-effect explanation, but fall short.

Data: Measure before/after scores for one group that receives a treatment and for a different group that does not receive the treatment (see Table 6.2b).

Example: The treatment may cause a reduction in smoking behavior but the reduced smoking may be caused by something else.

Nonexperimental

Purpose: Produce a description of the relationship between two variables but do not attempt to explain the relationship.

Data: Measure scores for two different groups of participants or for one group at two different times (see Table 6.2c).

Example: There is a relationship between gender and verbal ability. Girls tend to have higher verbal skills than boys, but we don't know why.

Case Study

Definition:

- A case study involves the **in-depth and detailed examination** of a single case, such as an individual, group, organization, or event.
- Focuses on understanding the **complexity and uniqueness** of the case within its real-life context.

Purpose:

- To gain **deep insights** into a specific phenomenon.
- Often used to explore **unique or critical cases** that provide valuable lessons or insights.

Common Use Cases:

- Used in **social sciences**, **business**, and **clinical studies**.
- Example: Studying a successful company to understand its strategies.

Types of Case Studies

Critical Case:

- Used to test a **clearly defined hypothesis**.
- Helps understand the conditions under which the hypothesis holds true.

Unique Case:

- Focuses on a **rare or unusual phenomenon**.
- Example: Margaret Mead's study of Samoan youth to understand their stress-free adolescence.

Revelatory Case:

- Examines a phenomenon that was previously **inaccessible to scientific investigation**.
- Provides new insights into unexplored areas.

Exemplifying Case:

- Chosen because it provides a **suitable context** for answering specific research questions.
- Helps in understanding **key social processes**.

Action Research

Action research is a **participatory and iterative process** where researchers and practitioners collaborate to solve real-world problems.

- Focuses on **improving practices** while simultaneously generating knowledge.

Key Features:

- **Cyclic Process**: Involves planning, acting, observing, and reflecting in repeated cycles.
- **Collaborative**: Researchers work closely with participants (e.g., teachers, employees, community members).
- **Problem-Solving Oriented**: Aims to address specific issues in a practical context.



Theoretical Research

Theoretical research focuses on **developing, exploring, or testing theories** to explain phenomena.

Theoretical research is a **logical exploration** of a system of beliefs and assumptions.

This type of research includes theorizing or defining how a cyber system and its environment behave and then exploring or playing out the implications of how it is defined.

Often theoretical research is decried as out of touch with reality, so-called "ivory tower" research.

In cyber security research, theoretical work often overlaps with mathematics, logic, or theory of computation; cryptography, of course, is a great example of this.

It aims to **expand knowledge** and understanding of concepts, principles, and relationships.

Key Features:

- **Conceptual Focus**: Emphasizes abstract ideas, models, and frameworks.
- **Hypothesis-Driven**: Often involves formulating and testing hypotheses to validate theories.
- **No Direct Data Collection**: Relies on existing literature, logical reasoning, and simulations.

Purpose:

- To **build or refine theories** that explain how and why certain phenomena occur.
- To provide a **foundation for empirical research**.

Simulation-based Research

Simulation based science is all about **formulating computational models of phenomena** that we observe and performing computer simulations in order to deepen our understanding of the systems that underpin these phenomena.

Simulation-based research involves creating **realistic, controlled environments** to study complex systems, behaviors, or interventions.

It uses **models, or virtual simulations** to replicate real-world scenarios.

Key Features:

- **Controlled Environment**: Allows researchers to manipulate variables and observe outcomes without real-world risks.
- **High-Fidelity Replication**: Uses advanced technology to mimic real-life conditions accurately.
- **Interdisciplinary Approach**: Combines fields like medicine, engineering, and social sciences.

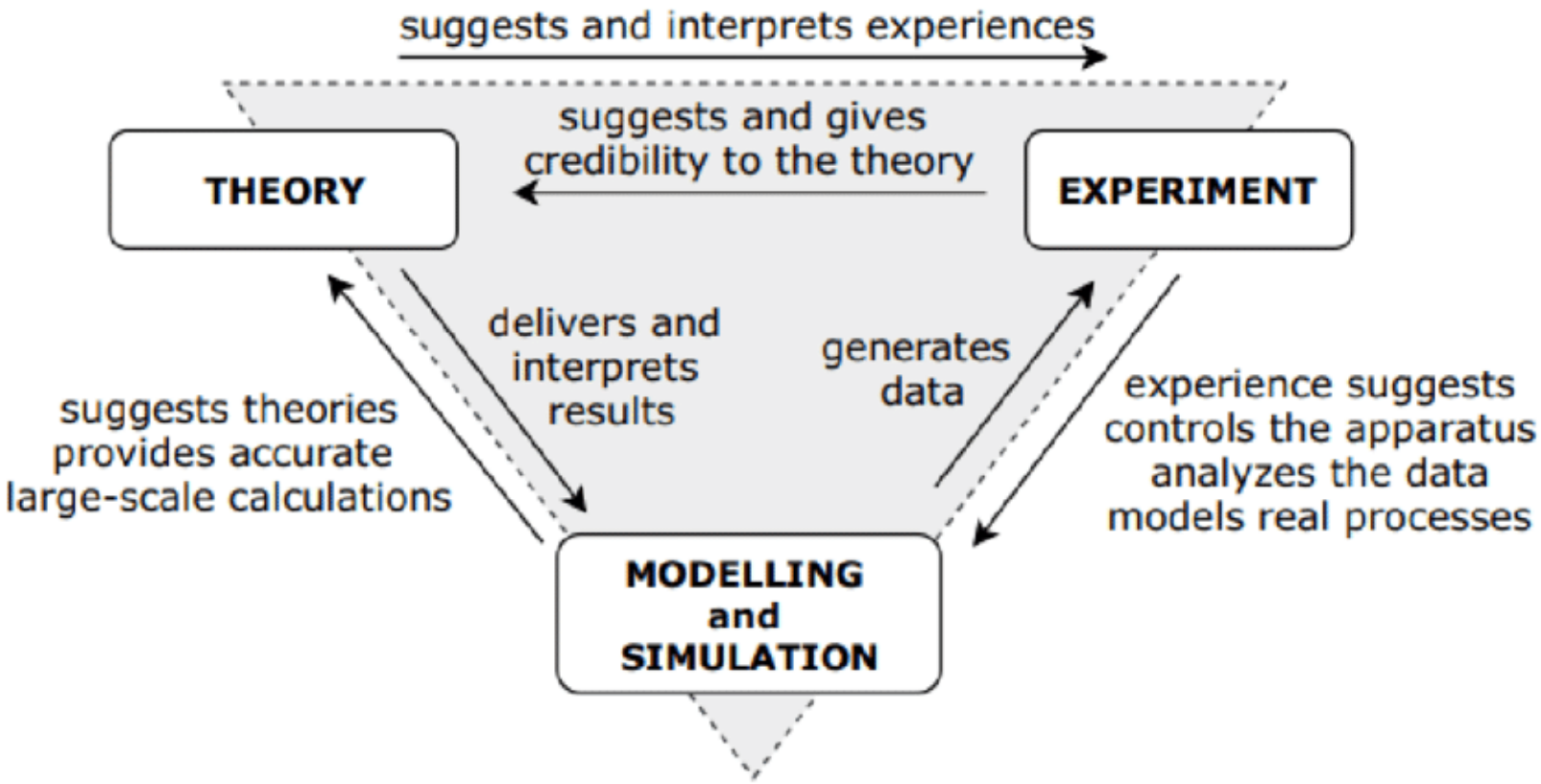
Applications:

- **Healthcare**: Training medical professionals, testing new treatments, and improving patient safety.
- **Engineering**: Simulating systems like autonomous vehicles or aerospace technologies.
- **Social Sciences**: Modeling human behavior or societal responses to policies.

Benefits:

- **Risk-Free Learning**: Enables practice and experimentation without real-world consequences.
- **Data-Driven Insights**: Provides measurable outcomes for analysis and improvement.
- **Cost-Effective**: Reduces the need for expensive real-world trials.

Simulation-based Research



Multidisciplinary Research Approaches

Multidisciplinary research involves integrating knowledge, methods, and perspectives from multiple disciplines to address complex problems.

Multidisciplinary research is “the linkage of phenomena, research approaches, and conceptual tools that had previously been pursued independently”. *Pickett et al, 1999: 302*

It encourages collaboration across fields to achieve comprehensive solutions.

Key Features:

- Collaborative Teams: Researchers from different disciplines work together.
- Diverse Perspectives: Combines insights from fields like science, engineering, social sciences, and humanities.
- Holistic Solutions: Addresses problems from multiple angles for more effective outcomes.

Applications:

- Healthcare: Combining medicine, engineering, and data science to develop advanced treatments.
- Environmental Studies: Integrating biology, chemistry, and policy-making to tackle climate change.
- Technology: Merging computer science, psychology, and design to create user-friendly systems.

Benefits:

- Innovation: Fosters creative solutions by blending diverse ideas.
- Comprehensive Understanding: Provides a deeper and more nuanced view of complex issues.
- Real-World Impact: Addresses societal challenges more effectively.

Example:

- A project combining biology, engineering, and ethics to develop sustainable agricultural practices.

Strategies for Multidisciplinary Research Teams

- Building the team
- Choosing the research question
- Developing the proposal
- Collecting the data
- Analyzing and making meaning of the data
- Writing, reporting and disseminating

Computational biology research



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